THE CHINESE UNIVERSITY OF HONG KONG, SHENZHEN
School of Management and Economics

ECONOMICS 3121
Introductory Econometrics

Spring term 2017 **MIDTERM EXAM** Yi Ding

DATE: **Sunday March 25, 2017**

TIME: **150 minutes; 3:00 p.m. – 5:30 p.m.**

INSTRUCTIONS: The exam consists of **FIVE (5)** questions. Students are required to answer **ALL FIVE (5)** questions.

Answer all questions in the exam booklets provided. Be sure your ***name*** and ***student number*** are printed clearly on the front of all exam booklets used.

Do not write answers to questions on the front page of the first exam booklet.

Please label clearly each of your answers in the exam booklets with the appropriate number and letter.

Please write legibly.

This exam is **CONFIDENTIAL**. This question paper must be submitted in its entirety with your answer booklet(s); otherwise your exam will not be marked.

A formula sheet and a table of percentage points of the t-distribution is given on the last pages of the exam.

MARKING: The marks for each question are indicated in parentheses immediately above each question. **Total marks** for the exam **equal 130**.

GOOD LUCK!

Question 1: Multiple Choice (40 points, 2 points each)
Select the **BEST** response to each of the following questions.

1. Which of the following correctly identifies a difference between cross-sectional data and time series data?
   a) Cross-sectional data is based on temporal ordering, whereas time series data is not.
   b) Time series data is based on temporal ordering, whereas cross-sectional data is not.
   c) Cross-sectional data consists of only qualitative variables, whereas time series data consists of only quantitative variables.
   d) Time series data consists of only qualitative variables, whereas cross-sectional data does not include qualitative variables.

Answer: b

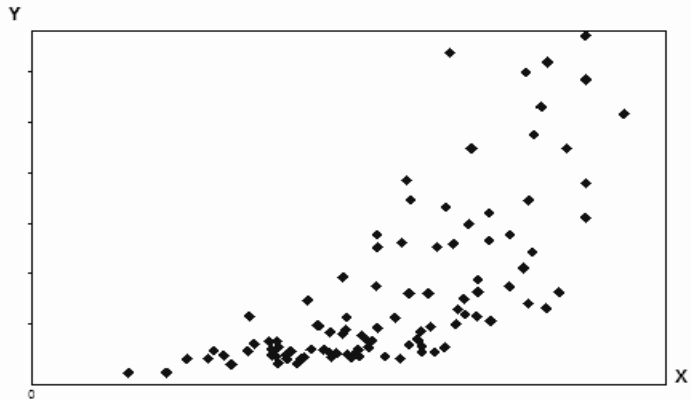
2. Which of the following is NOT correct with regard to the *p*-value attached to a test statistic?
   a) *p*-values can only be used for two-sided tests
   b) It is the marginal significance level where we would be indifferent between rejecting and not rejecting the null hypothesis
   c) It is the exact significance level for the test
   d) Given the *p*-value, we can make inferences without referring to statistical tables

Answer: a

3. The slope estimator, $\hat{\beta}_1$, has a larger standard error, other things equal, if
   a) there is more variation in explanatory variable, x.
   b) there is a smaller variance of error term, u.
   c) the sample size is smaller.
   d) the intercept estimator, $\hat{\beta}_0$, is larger.

Answer: c

4. Which regression model would best fit the data shown in the plot below?
   a) $y = \beta_0 + \beta_1 x + u$
   b) $y = \beta_0 + \beta_1 \log(x) + u$
   c) $\log(y) = \beta_0 + \beta_1 x + u$
   d) $\log(y) = \beta_0 + \beta_1 \log(x) + u$



Answer: c

5. Which of the following statements is true when the dependent variable, $y > 0$
   a) Taking log of a variable often expands its range.
   b) Models using $log(y)$ as the dependent variable may satisfy CLM assumptions more closely than models using the level of y.

c) Taking log of variables make OLS estimates more sensitive to extreme values.
d) Taking logarithmic form of variables make the slope coefficients more responsive to rescaling.

Answer: b

6. In a multiple OLS regression, if you divide the explained variable y by 12 (to change units of measurement from inches to feet) which of the following will happen?
a) The estimated intercept parameter will be reduced by 12 times.
b) the estimated slope parameter will be reduced by 12 units.
c) the total sum of squares for the regression will be reduced by $\sqrt{12}$ times
d) R-squared for the regression will be reduced by 12 times.

Answer: a

7. A researcher has just finished a statistical analysis and claims that he found evidence that x affects y positively. Using a 1-tailed test, he found that the estimated coefficient for x is significantly positive at the 5% level, but is not significant at the 4% level. Assume that the classical linear regression assumptions hold. If the researcher used a 2-tailed test instead, what would he have found?
a) The estimated coefficient for x is significantly positive at the 10% level.
b) The estimated coefficient for x is significantly positive at the 5% level.
c) The estimated coefficient for x is significantly positive at the 4% level.
d) The estimated coefficient for x is significantly positive at the 2.5% level.

Answer: a

8. In a multiple OLS regression, what is the consequence of including an irrelevant variable?
a) The OLS estimators are biased.
b) The R-square of the regression may decrease.
c) The t statistic the OLS estimators may increase
d) The estimated variance of the OLS estimators may increase.

Answer: d

9. Which of the following statements is true?
a) If the calculated value of F statistic is higher than the critical value, we reject the alternative hypothesis in favor of the null hypothesis.
b) The F statistic is always nonnegative as SSR_r is never smaller than SSR_{ur}.
c) Degrees of freedom of a restricted model is always less than the degrees of freedom of an unrestricted model.
d) The F statistic is more flexible than the t statistic to test a hypothesis with a single restriction.

Answer: b

10. Suppose that the value of R^2 for an estimated regression model is exactly one. Which of the following are true?
(i) All of the data points must lie exactly on the line
(ii) All of the residuals must be zero

(iii) All of the variability of *y* have been explained by the model
(iv) The fitted line will be horizontal with respect to all of the explanatory variables

a) (ii) and (iv) only
b) (i) and (iii) only
c) (i), (ii), and (iii) only
d) (i), (ii), (iii), and (iv)

Answer: c

11. In a regression model where $y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + u$, to test the hypothesis that $\beta_1 - 2\beta_2 = 0$ we should:

a) Regress y on $x_1, (x_1 + 2x_2), x_3$ and examine the significance of the coefficient for x_1
b) Regress y on $x_1, (x_1 + 2x_2), x_3$ and examine the significance of the coefficient for $(x_1 + 2x_2)$
c) Regress y on $x_1, (2x_1 + x_2), x_3$ and examine the significance of the coefficient for x_1
d) Regress y on $x_1, (2x_1 + x_2), x_3$ and examine the significance of the coefficient for $(2x_1 + x_2)$

Answer: c

12. Consider the following regression model on 64 observations:

$y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \beta_3 x_{i3} + \beta_4 x_{i4} + u_i$

Which of the following hypotheses could we test using an *F*-test?

(i) $H_0: \beta_2 = 0$ $\quad H_1: \beta_2 < 0$
(ii) $H_0: \beta_2 = 1 \; and \; \beta_3 + \beta_4 = 1$ $\quad H_1: H_0 \; is \; not \; true$
(iii) $H_0: \beta_3 \beta_4 = 1$ $\quad H_1: \beta_3 \beta_4 \neq 1$
(iv) $H_0: \beta_2 - \beta_3 - \beta_4 = 1$ $\quad H_1: \beta_2 - \beta_3 - \beta_4 \neq 1$

a) (ii) only
b) (ii) and (iv) only
c) (i), (ii), (iii) and (iv)
d) (i), (ii), and (iv) only

Answer: b

13. You have just read a paper which claims that there is strong evidence that x affects y negatively. The researcher found that the estimated coefficient for x is significantly negative at the 5% level. You suspect that there is omitted variable bias. Which of the following statements is correct?

a) If there is omitted variable bias, and the bias is positive, then the researcher's conclusions are invalid.
b) If there is omitted variable bias, and the bias is positive, then the researcher's conclusions are still valid.
c) If there is omitted variable bias, and the bias is zero, then the researcher's conclusions are invalid.
d) If there is omitted variable bias, and the bias is negative, then the researcher's conclusions are still valid.

Answer: a

14. If the estimates of the coefficients of interest change substantially across specifications,
    a) then this can be expected from sample variation.
    b) then you should change the scale of the variables to make the changes appear to be smaller.
    c) then this often provides evidence that the original specification had omitted variable bias.
    d) then choose the specification for which your coefficient of interest is most significant.

Answer: c

You have been commissioned to construct a model that explains NBA players' wages. The dependent variable is $lwage$, the natural log of players' annual wages (*in thousands of dollors*). Please use the following Stata regression results for the next **SIX** questions.

Source	SS	df	MS
Model	90.2698185	5	18.0539637
Residual	117.918945	263	.448361006
Total	208.188763	268	.776823743

Number of obs	= 269
F(5, 263)	= 40.27
Prob > F	= 0.0000
R-squared	= 0.4336
Adj R-squared	= 0.4228
Root MSE	= .6696

lwage	Coef.	Std. Err.	t	P>|t|	[95% Conf.	Interval]
games	.0004132	.002682	0.15	0.878	-.0048679	.0056942
avgmin	.0302278	.0130868	2.31	0.022	.0044597	.055996
points	.0363734	.0150945	2.41	0.017	.0066519	.0660949
rebounds	.0406795	.0229455	1.77	0.077	-.0045007	.0858597
turnovers	-.0366593		-2.65	0.008	-.0637731	
_cons	5.648996	.1559075	36.23	0.000	5.34201	5.955982

15. According to the regression results above, what is the F statistic that simultaneously tests the restrictions that all slope parameters equal to 0.
    a) 0.4336
    b) 40.27
    c) 269
    d) 0.0000

Answer: b

16. According to the regression results above, what is the standard error of regression?
    a) 0.6696
    b) 0.448361006
    c) 117.918945
    d) 18.0539637

Answer: a

17. In the above regression, which of the variables are significant at the 10% level?
    a) $games,\ avgmin,\ points,\ rebounds,\ turnovers$
    b) $avgmin,\ points,\ rebounds,\ turnovers$

c) $avgmin,\ points,\ turnovers$
d) $points$

Answer: b

18. In the above regression, which of the variables show a significant **positive** effect at the 1% level?

a) $points,\ turnovers$
b) $points$
c) $games$
d) None of the variables are positively significant at the 1% level

Answer: b

19. According to the above regression results, what is the expected change in wages when $points$ changes, all else remaining constant?

a) Wages will increase by $36.37 when $points$ increases by 1 unit.
b) Wages will increase by 0.03637% when $points$ increases by 1%.
c) Wages will increase by $0.03637 when $points$ increases by 1 unit.
d) Wages will increase by 3.637% when $points$ increases by 1 unit.

Answer: d

20. In the above regression, what is the upper bound of 95% confidence interval of the slope parameter on $turnovers$?

a) -0.0095455
b) 0.260384
c) -0.0271138
d) -0.0637731

Answer: a

Question 2 (20 marks)

Discuss the effect of the following phenomena on the unbiasedness and the variance of the OLS estimators, $\hat{\beta}_j$:

1. Including an irrelevant variable

Effect on the unbiasedness of $\hat{\beta}_j$: Including an irrelevant variable does not affect the unbiasedness of the OLS estimators, $\hat{\beta}_j$. **(2 marks)**

Effect on the variance of $\hat{\beta}_j$: Including an irrelevant variable increases the variances of the OLS estimators of the relevant independent variables. In particular, R_j^2 increases and thus, the variance of $\hat{\beta}_j$, given by $var(\hat{\beta}_j) = \frac{\sigma^2}{SST_j(1-R_j^2)}$, increases. **(2 marks)**

2. Omitting a relevant variable

Effect on the unbiasedness of $\hat{\beta}_j$: If the omitted relevant variable is correlated with some of the other independent variables in the model, omitting the relevant variable biases the OLS estimators, $\hat{\beta}_j$. In particular, the omitted relevant variable will be included in the error term u, and will violate the zero conditional mean assumption (Assumption 4) **(1 marks)**. If the omitted relevant variable is not correlated with any of the other independent variables in the model, omitting the relevant variable will not bias the OLS estimators, $\hat{\beta}_j$. **(1 marks)**

Effect on the variance of $\hat{\beta}_j$: Omitting a relevant variable decreases the variances of the OLS estimators of the remaining independent variables. In particular, R_j^2 decreases and thus, the variance of $\hat{\beta}_j$, given by $var(\hat{\beta}_j) = \frac{\sigma^2}{SST_j(1-R_j^2)}$, decreases. (**Or:** We assume that variances of error term, σ^2, doesn't change when omitting a relevant variable. In fact, σ^2 increase when omitting a relevant variable, so both the denominator and numerator of variance of $\hat{\beta}_j$ increase and therefore, the variance of $\hat{\beta}_j$ can increase, decrease, or remain unchanged) **(2 marks)**

3. Perfect collinearity

Perfect collinearity means that some independent variable, x_j, is a perfect linear combination of other independent variables.

Effect on the unbiasedness of $\hat{\beta}_j$: Perfect collinearity violates Assumption 3 for the unbiasedness of OLS estimators, $\hat{\beta}_j$. In fact, under perfect collinearity, the OLS estimators, $\hat{\beta}_j$, practically cannot be computed. That's why we have to drop one of the variables in the perfect linear combination. **(2 marks)**

Effect on the variance of $\hat{\beta}_j$: If the independant variable,x_j, is perfect correlated with other independent variables, then $R_j^2 = 1$ and the variance of $\hat{\beta}_j$, given by $var(\hat{\beta}_j) = \frac{\sigma^2}{SST_j(1-R_j^2)}$, is not defined. **(2 marks)**

4. Multicollinearity

Multicollinearity means that some independent variables are highly correlated with each other.

Effect on the unbiasedness of $\hat{\beta}_j$: Multicollinearity does not affect the unbiasedness of the OLS estimators, $\hat{\beta}_j$. **(2 marks)**

Effect on the variance of $\hat{\beta}_j$: If the independant variable,x_j, is highly correlated with other independent variables, then R_j^2 is "close to 1". Thus, the variance of $\hat{\beta}_j$, given by $var(\hat{\beta}_j) = \frac{\sigma^2}{SST_j(1-R_j^2)}$, is large and this decreases the precision of $\hat{\beta}_j$. **(2 marks)**

5. Heteroskedasticity

Heteroskedasticity means that the error variance is not a constant and so Assumption 5 of the Gauss-Markov assumptions is violated, that is $var(u|x) \neq \sigma^2$

Effect on the unbiasedness of $\hat{\beta}_j$: Heteroskedasticity does not affect the unbiasedness of the OLS estimators, $\hat{\beta}_j$. **(2 marks)**

Effect on the variance of $\hat{\beta}_j$: Under heteroskedasticity, the usual formula for the variance of OLS estimators, $(\hat{\beta}_j) = \frac{\sigma^2}{SST_j(1-R_j^2)}$, is no longer valid. **(2 marks)**

Question 3 (20 marks)
Answer parts (a), (b), and (c) below

(4 marks)
(a) Consider the OLS regression function, $\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x_1 + \hat{\beta}_2 x_2$. Briefly explain how to use partialling out interpretation of multiple regression to find the OLS estimator $\hat{\beta}_1$.

The partialling out interpretation involves two steps:

1) Do a sample regression of x_1 on x_2 to obtain the residual $\hat{r}_1$ where the effects of x_2 have been partialled out. **(2 marks)**

2) Regress y on residual $\hat{r}_1$, and the slope coefficient represents ceteris paribus effect of x_1 on y. It can be expressed as:

$$\hat{\beta}_j = \frac{\sum(\hat{r}_{i1} y_i)}{\sum \hat{r}_{i1}{}^2}$$ **(2 marks)**

(8 marks)
(b) Using the formula of OLS estimator $\hat{\beta}_1$ derived in part (a), prove that $\hat{\beta}_1$is an unbiased estimator of the slope coefficient β_1. Be specific.

$$\hat{\beta}_j = \frac{\sum(\hat{r}_{i1} y_i)}{\sum \hat{r}_{i1}^2}$$ Replace y_i with $\beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + u_i$

$$= \frac{\sum(\hat{r}_{i1}(\beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + u_i))}{\sum \hat{r}_{i1}^2}$$

$$= \frac{\beta_0 \sum \hat{r}_{i1}}{\sum \hat{r}_{i1}^2} + \frac{\beta_1 \sum \hat{r}_{i1} x_{i1}}{\sum \hat{r}_{i1}^2} + \frac{\beta_2 \sum \hat{r}_{i1} x_{i2}}{\sum \hat{r}_{i1}^2} + \frac{\sum \hat{r}_{i1} u_i}{\sum \hat{r}_{i1}^2}$$ Since $\sum \hat{r}_{i1}$=0, $\sum \hat{r}_{i1} x_{i2} = 0$, and $\sum \hat{r}_{i1} x_{i1}$=$\sum \hat{r}_{i1}^2$

$$= \beta_1 + \frac{\sum(\hat{r}_{i1} u_i)}{\sum \hat{r}_{i1}^2}$$ **(4 marks)**

Take expectation on $\hat{\beta}_j$

$$E(\hat{\beta}_j) = \beta_1 + E\left[\frac{\sum(\hat{r}_{i1} u_i)}{\sum \hat{r}_{i1}^2}\right]$$ Since $\hat{r}_{i1}$ is a function of X only, $\hat{r}_{i1}$ is fixed in repeated samples

$$= \beta_1 + \frac{\sum(\hat{r}_{i1} E(u_i))}{\sum \hat{r}_{i1}^2}$$ Since $E(u_i) = 0$

$$= \beta_1$$

Therefore, $\hat{\beta}_1$is an unbiased estimator of the slope coefficient β_1. **(4 marks)**

(8 marks)

(c) Suppose the true population model is $y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + u$. Prove that the OLS estimator $\hat{\beta}_1$ in part (a) is biased. Be specific.

$$\hat{\beta}_j = \frac{\sum(\hat{r}_{i1} y_i)}{\sum \hat{r}_{i1}^2}$$ Replace y_i with $\beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \beta_3 x_{i3} + u_i$

$$= \frac{\sum(\hat{r}_{i1}(\beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \beta_3 x_{i3} + u_i))}{\sum \hat{r}_{i1}^2}$$

$$= \frac{\beta_0 \sum \hat{r}_{i1}}{\sum \hat{r}_{i1}^2} + \frac{\beta_1 \sum \hat{r}_{i1} x_{i1}}{\sum \hat{r}_{i1}^2} + \frac{\beta_2 \sum \hat{r}_{i1} x_{i2}}{\sum \hat{r}_{i1}^2} + \frac{\beta_3 \sum \hat{r}_{i1} x_{i3}}{\sum \hat{r}_{i1}^2} + \frac{\sum \hat{r}_{i1} u_i}{\sum \hat{r}_{i1}^2}$$ Since $\sum \hat{r}_{i1} = 0$, $\sum \hat{r}_{i1} x_{i2} = 0$, and $\sum \hat{r}_{i1} x_{i1} = \sum \hat{r}_{i1}^2$

$$= \beta_1 + \frac{\beta_3 \sum \hat{r}_{i1} x_{i3}}{\sum \hat{r}_{i1}^2} + \frac{\sum(\hat{r}_{i1} u_i)}{\sum \hat{r}_{i1}^2}$$ **(4 marks)**

Take expectation on $\hat{\beta}_j$

$$E(\hat{\beta}_j) = \beta_1 + E\left[\frac{\beta_3 \sum \hat{r}_{i1} x_{i3}}{\sum \hat{r}_{i1}^2}\right] + E\left[\frac{\sum(\hat{r}_{i1} u_i)}{\sum \hat{r}_{i1}^2}\right]$$ Since $\hat{r}_{i1}$ is a function of X only, $\hat{r}_{i1}$ is fixed in repeated samples

$$= \beta_1 + \frac{\beta_3 \sum \hat{r}_{i1} x_{i3}}{\sum \hat{r}_{i1}^2} + \frac{\sum(\hat{r}_{i1} E(u_i))}{\sum \hat{r}_{i1}^2}$$ Since $E(u_i) = 0$

$$= \beta_1 + \frac{\beta_3 \sum \hat{r}_{i1} x_{i3}}{\sum \hat{r}_{i1}^2}$$

Therefore, $\hat{\beta}_1$ is a biased estimator of the slope coefficient β_1. **(4 marks)**

Question 4 (20 points)

You have been commissioned to investigate the relationship between the annual salaries of Chief Executive Officers (CEOs) of firms and the annual profits of their firms. The dependent variable is $\boldsymbol{salary_i}$, the annual salary of the CEO of the i-th firm, measured in *thousands of dollars* per year. The explanatory variable is $\boldsymbol{profits_i}$, the annual profits of the i-th firm, measured in *millions of dollars per year*. The model you propose to estimate is given by the population regression equation

$$salary_i = \beta_0 + \beta_1 profits_i + u_i$$

Your research assistant has used observations on $\boldsymbol{salary_i}$ and $\boldsymbol{profits_i}$ for a sample of 177 corporations to estimate the following OLS sample regression equation, where the figures in parentheses below the coefficient estimates are the *standard errors* of the coefficient estimates:

$$\underset{\quad\quad\quad

$\widehat{\beta_1} - c_\alpha se(\widehat{\beta_1}) \leq \beta_1 \leq \widehat{\beta_1} + c_\alpha se(\widehat{\beta_1})$ **(2 marks)**

Degree of freedom $= n - 2 = 177 - 2 = 175$
Critical value of t_{175} at 5% significance level ($\alpha = 0.01$) is $\boldsymbol{c_{0.05} = 1.960}$

$c_\alpha se(\widehat{\beta_1}) = 1.960 * 0.10094 = 0.1978$

Lower 95% confidence limit for β_1 is: **(2 marks)**

$\widehat{\beta_1} - c_\alpha se(\widehat{\beta_1}) = 0.5723 - 0.1978 = \mathbf{0.3745}$

Upper 95% confidence limit for β_1 is: **(2 marks)**

$\widehat{\beta_1} + c_\alpha se(\widehat{\beta_1}) = 0.5723 + 0.1978 = \mathbf{0.7701}$

Result: The two-sided 95% confidence interval for β_1 is: [0.3745, 0.7701]

(7 points)
(c) Perform a test of the proposition that an increase in profits of 1 million dollars per year is associated on average with an increase in CEO salary of 500 dollars per year. Use the 5 percent significance level (i.e., $\alpha = 0.05$). State the null hypothesis H_0 and the alternative hypothesis H_1. Show how you calculated the test statistic. State the decision rule you use, and the inference you would draw from the test.

Null hypothesis: $H_0: \beta_1 = 0.5$
Alternative hypothesis: $H_1: \beta_1 \neq 0.5$ a ***two-tail*** **test** **(1 marks)**

t-test statistic under H_0 is: **(3 marks)**

$$t(\widehat{\beta_1}) = \frac{\widehat{\beta_1} - \beta_1}{se(\widehat{\beta_1})} = \frac{\widehat{\beta_1} - 0.5}{se(\widehat{\beta_1})} = \frac{0.57230 - 0.5}{0.10094} = \mathbf{0.716}$$

Decision Rule: This is a ***two-tail*** **test**. Compare the sample t statistic with the α-level critical value of the t_{175} distribution. **(1 marks)**
1. If $|t| \geq c_\alpha$, do not reject H_0 at the significance level α.
2. If $|t| < c_\alpha$, reject H_0 at the significance level α.

Degree of freedom $= n - 2 = 177 - 2 = 175$
Critical value of t_{175} at 1% significance level ($\alpha = 0.05$) is $\boldsymbol{c_{0.05} = 1.960}$ **(1 marks)**

Inference: **(1 marks)**
Since $\boldsymbol{t = 0.716 < 1.960 = c_{0.05}}$**, do not reject $\mathbf{H_0}$** at the **5%** significance level.

Question 5 (30 marks)

You are conducting an econometric investigation into the effect on house prices of proximity to a power plant, which presumably generates negative externalities for homeowners and others located close to it. The sample data consist of observations for 321 houses that were sold in a single metropolitan area in the year 2012 on the following variables:

$price_i$: selling price of house i, in dollars.
$hsize_i$: living area of house i, in square meters.
age_i : age of house i, in years.
$dist_i$: distance of house i from power plant, in meters.

The regression model you propose to use is as follows:

$$price_i = \beta_0 + \beta_1 hsize_i + \beta_2 age_i + \beta_3 age_i^2 + \beta_4 dist_i + \beta_5 dist_i^2 + \beta_6 age_i dist_i + u_i \quad (1)$$

Using the sample data described above, your research assistant computes OLS estimates of regression equation (1) and of two restricted versions of equation (1). For each of the sample regression equations estimated on the sample of n = 321 observations, the following table contains the OLS coefficient estimates (with estimated standard errors in parentheses below the coefficient estimates), R-squared value, and number of sample observations.

Dependent variable: $price_i$				
Independent variables		**(1)**	**(2)**	**(3)**
$hsize_i$	$\widehat{\beta_1}$	363.2 (28.65)	361.1 (28.54)	373.1 (28.61)
age_i	$\widehat{\beta_2}$	-1,005 (192.4)	-915.7 (163.5)	-923.5 (140.1)
age_i^2	$\widehat{\beta_3}$	3.496 (1.096)	3.743 (1.059)	3.825 (0.957)
$dist_i$	$\widehat{\beta_4}$	7.400 (3.582)	8.220 (3.456)	
$dist_i^2$	$\widehat{\beta_5}$	-0.000656 (0.000257)	-0.000695 (0.000254)	
$age_i dist_i$	$\widehat{\beta_6}$	0.0232 (0.0265)		
$Intercept$	$\widehat{\beta_0}$	20,168 (13,130)	17,222 (12,689)	34,426 (6,435)
Observations		321	321	321
R-squared		0.5466	0.5455	0.5318

(10 marks)

(a) State the coefficient restriction(s) that regression equation (3) imposes on regression equation (1). Explain in words what the restriction(s) mean. Use the results given above from OLS estimation of regression equations (1) and (3) to perform a test of these coefficient restrictions at the 5 percent significance level (i.e., for significance level $\alpha = 0.05$). State the null and alternative hypotheses, show how you calculate the required test statistic. State the decision rule you use, and the inference you would draw from the test. Based on the outcome of the test, which of the two regression equations would you choose, equation (1) or equation (3)?

Equation (3) imposes on equation (1) the exclusion restrictions $\boldsymbol{\beta_4 = 0, \beta_5 = 0}$ **and** $\boldsymbol{\beta_6 = 0}$ **(1 marks)**

Test: $H_0: \beta_4 = 0, \beta_5 = 0, \beta_6 = 0$ versus: H_1: H_0 is not true **(1 marks)**

Interpretation of $\boldsymbol{H_0}$: Distance from power plant, the quadratic term of distance from power plant, and the interaction term between distance from power plant and age of house do not affect the housing price. (**Or:** the partial effect of distance from power plant on housing price is zero) **(1 marks)**

Compute F statistic: **(3 marks)**

$$F = \frac{(R_{ur}^2 - R_r^2)/q}{(1 - R_{ur}^2)/(n - k - 1)} = \frac{(0.5466 - 0.5318)/3}{(1 - 0.5466)/(321 - 6 - 1)} = \mathbf{3.4166}$$

Decision Rule: Compare the sample F statistic with the α-level critical value of the $\mathbf{F_{3,314}}$ distribution. **(1 marks)**

1. If $F \leq c_\alpha$, do not reject H_0 at the significance level α.
2. If $F > c_\alpha$, reject H_0 at the significance level α.

Critical value of $\mathbf{F_{3,314}}$ at 5% significance level ($\alpha = 0.05$) is $\boldsymbol{c_{0.05} = 2.60}$ **(1 marks)**

Inference: **(1 marks)**

Since $\boldsymbol{F = 3.4166 > 2.60 = c_{0.05}}$**, reject $\mathbf{H_0}$** at the **5%** significance level.

Choose equation (1): restrictions incorporated in equation (3) are rejected at a sufficiently low significance level. **(1 marks)**

(10 marks)

(b) Write the expression for the marginal effect of age_i on $price_i$ implied by regression equation (1). Compute a test of the null hypothesis that the marginal effect of age_i on $price_i$ was constant for all houses in regression equation (1). Perform the test at the 5 percent significance level (i.e., for significance level $\alpha = 0.05$). State the null hypothesis H_0 and the alternative hypothesis H_1. Write the restricted regression equation implied by the null hypothesis H_0. OLS estimation of this restricted regression equation yields an R-squared value or $R^2 = 0.5274$. Use this information, together with the results from OLS estimation of equation (1), to calculate the required test statistic. State the decision rule you use, and the inference you would draw from the test.

The **marginal effect of** age_i on $price_i$ is:

$\frac{\partial price_i}{\partial age_i} = \beta_2 + 2\beta_3 age_i + \beta_6 dist_i$ **(1 marks)**

Null hypothesis: $H_0: \beta_3 = 0, \beta_6 = 0$
Alternative hypothesis: H_1: H_0 is not true **(2 marks)**

Restricted model implied by the null hypothesis: **(1 marks)**
$price_i = \alpha_0 + \alpha_1 hsize_i + \alpha_2 age_i + \alpha_3 dist_i + \alpha_4 dist_i^2 + u_i$

Compute F statistic: **(3 marks)**
$$F = \frac{(R_{ur}^2 - R_r^2)/q}{(1 - R_{ur}^2)/(n-k-1)} = \frac{(0.5466 - 0.5274)/2}{(1-0.5466)/(321-6-1)} = \mathbf{6.6484}$$

Decision Rule: Compare the sample F statistic with the α-level critical value of the $\mathbf{F_{2,314}}$ distribution. **(1 marks)**
1. If $F \leq c_\alpha$, do not reject H_0 at the significance level α.
2. If $F > c_\alpha$, reject H_0 at the significance level α.

Critical value of $\mathbf{F_{2,314}}$ at 5% significance level ($\alpha = 0.05$) is $\boldsymbol{c_{0.05} = 3.00}$ **(1 marks)**

Inference: **(1 marks)**
Since $\boldsymbol{F = 6.6484 > 3.00 = c_{0.05}}$**, reject $\mathbf{H_0}$** at the **5%** significance level.

(10 marks)
(c) Use the estimation results for regression equation (2) to perform a two-tail test of the null hypothesis $\beta_2 = -3\beta_1$ at the 5 percent significance level. Explain in words what this null hypothesis means. The estimated covariance of $\hat{\beta}_1$ and $\hat{\beta}_2$ is $\widehat{Cov}(\hat{\beta}_1, \hat{\beta}_2) = 1418.9597$. State the null and alternative hypotheses, show how you calculate the required test statistic, and state its null distribution. State the decision rule you use, and the inference you would draw from the test at the 5 percent significance level.

Null hypothesis: $H_0: 3\beta_1 + \beta_2 = 0$
Alternative hypothesis: H_1: H_0 is not true **(1 marks)**

Interpretation of $\boldsymbol{H_0}$: Decreasing the living area of house by 3 square meters and increasing the age of house by 1 year have the same ceteris paribus effect on housing price. **(1 marks)**

Compute t-statistic:
$3\hat{\beta}_1 + \hat{\beta}_2 = 3 * 361.1 - 915.7 = \mathbf{167.6}$ **(2 marks)**

$$\hat{Var}(3\hat{\beta}_1 + \hat{\beta}_2) = 9\hat{Var}(\hat{\beta}_1) + \hat{Var}(\hat{\beta}_2) + 6cov(\hat{\beta}_1, \hat{\beta}_2)$$
$$\hat{Var}(3\hat{\beta}_1 + \hat{\beta}_2) = 9(28.54)^2 + (163.5)^2 + 6(1418.9597)$$
$$\hat{Var}(3\hat{\beta}_1 + \hat{\beta}_2) = \mathbf{42576.7926}$$
$$se(3\hat{\beta}_1 + \hat{\beta}_2) = \sqrt{\hat{Var}(3\hat{\beta}_1 + \hat{\beta}_2)} = \mathbf{206.3414}$$ **(2 marks)**
$$t = \frac{3\hat{\beta}_1 + \hat{\beta}_2}{se(3\hat{\beta}_1 + \hat{\beta}_2)} = \frac{167.6}{206.3414} = \mathbf{0.8122}$$

(1 marks)

Decision Rule: This is a ***two-tail* test**. Compare the sample t statistic with the α-level critical value of the t_{314} distribution. **(1 mark)**

1. If $|t| \leq c_\alpha$, do not reject H_0 at the significance level α.
2. If $|t| > c_\alpha$, reject H_0 at the significance level α.

Critical value of t_{314} at 5% significance level ($\alpha = 0.05$) is $\boldsymbol{c_{0.05} = 1.96}$ **(1 mark)**

Inference: **(1 mark)**

Since $\boldsymbol{t = 0.8122 < 1.96 = c_{0.05}}$**, $\mathbf{H_0}$ cannot be rejected** at the **5%** significance level.